GRADUATE SCHOOL SOLAR ACADEMY

UNIVERSITÉ SAVOIE MONT BLANC

cnrs

Operational research for urban solar development

"Regional PV production modelling from open data"



Monday 19/10/2026 Session 1

Mohamed Ennhiri

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Curriculum

Plan

Day	Time	Duration	Content
Session 1 **Monday 19/10**	**8h00 - 9h30**	**1 h 30**	**Lecture 70 % / Hands-on 30 %**
Session 2 Tuesday 20/10	13h15 - 14h45	1 h 30	Lecture 30 % / Hands-on 70 %
Session 3 Wednesday 21/10 (TP 1) Thursday 22/10 (TP 2)	8h15 - 11h15 (TP 1) 13h00 - 16h00 (TP 2)	3 h	Lecture 15 % / Hands-on 60 % / Project 25 %
Session 4 Monday 09/11 (TP 1) Tuesday 10/11 (TP 2)	8h15 - 11h15 (TP 1) 8h15 - 11h15 (TP 2)	3 h	Lecture 10 % / Project 90 %
Session 5 Tuesday 24/11 (TP 2) Friday 27/11 (TP 1)	9h45 - 12h45 (TP 2) 13h15 - 16h15 (TP 1)	3 h	Project 100 %, then oral presentations
Total		12 h	

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Curriculum

Content

PV performance modeling from weather variables and system configuration, which includes:

- Plane-of-array irradiance
- Module temperature
- DC power, AC power
- Performance metrics

Fleet modeling: from one system to the installed capacity of a whole region, one weather node per department.

Validation against the production published by the grid operator, and error attribution.

Hands-on with python notebooks.

Structure inspired by the course of Alexandre Mathieu, Solar Academy 2024: PV failure detection based on operational time series.

Curriculum

Motivations

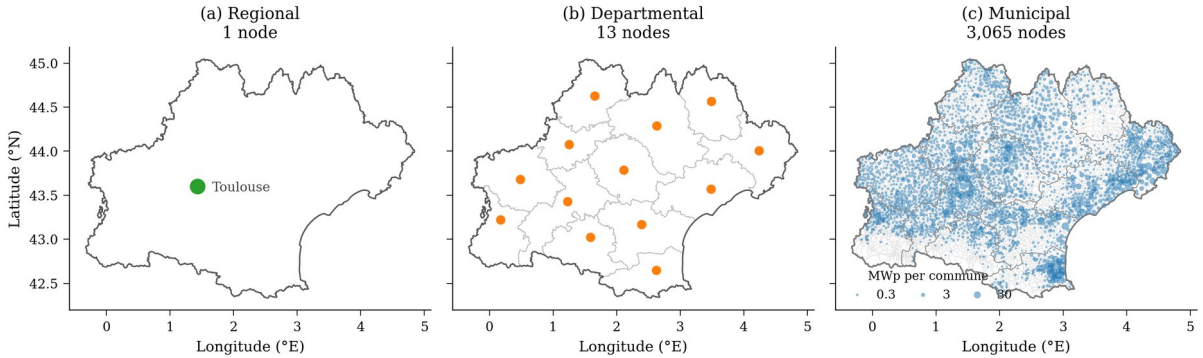
Distributed PV grows fast: thousands of small rooftop systems per department.

The grid operator sees the aggregated production, not the systems.

Capacity registers lag commissioning by weeks to months.

RED III: 49 % renewable share in buildings by 2030 needs validated, spatially resolved estimates.

Uses: nowcasting, curtailment studies, self-consumption sizing, register checks.



Ennhiri et al., EU PVSEC 2026. Occitanie, 3.7 GWp, 2023, three aggregation levels.

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Curriculum

Objectives

At the end of this course, you should be able to:

- Perform data treatment and analysis with Python on hourly weather and production time series.
- Model and calculate the performance of a PV installation.
- Scale a single-system model to a regional fleet weighted by installed capacity.
- Compare the fleet model with the reference production and attribute the gap to a cause.

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Final project

The project aims at estimating the hourly PV production of one region.

2 parts:

- A. Individual, student number n: estimation of the operating variables of one system, POA irradiance, module temperature, DC and AC power, compared with its actual data.
- B. In pairs, one region each: installed capacity register, a first model, a better representation of the fleet, comparison with the RTE series, error attribution.

Deliverables: one notebook per pair as html, csv outputs, one oral presentation per pair.

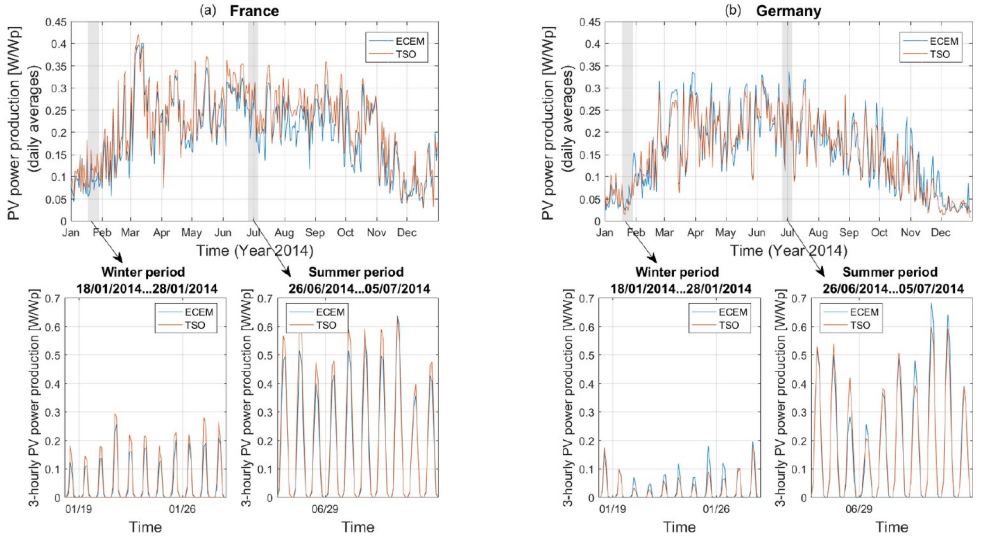


Figure: the same exercise for France and Germany, fleet model (blue) against TSO (red), 2014. Saint-Drenan, Wald, Ranchin, Dubus, Troccoli (2018), Adv. Sci. Res. 15, 51–62, doi:10.5194/asr-15-51-2018, CC BY 4.0.

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Audience

Background

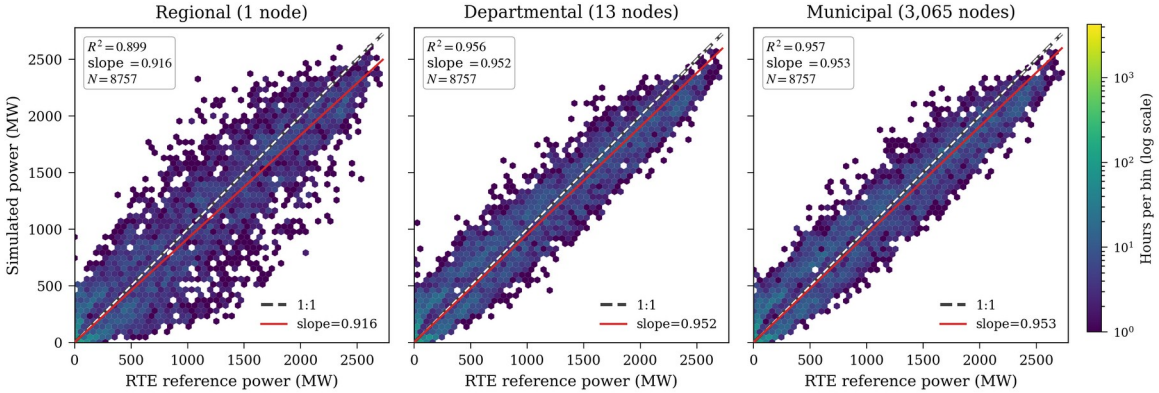
Seven questions, two minutes, no name: background, PV, Python, tools, programming languages, PV software, comfort with coding.

Who am I ?

PhD: regional estimation of distributed PV production from open data, top-down (physical fleet model) and bottom-up (per-rooftop database from aerial imagery and LiDAR).

LOCIE, UMR CNRS 5271, Université Savoie Mont Blanc, with LISTIC. Started January 2026.

First result: Occitanie, 3.7 GWp, simulated at 1, 13 and 3 065 nodes against the TSO hourly series. Departmental level: hourly R^2 0.957, annual energy within 0.5 %.



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Agenda

Curriculum

Audience

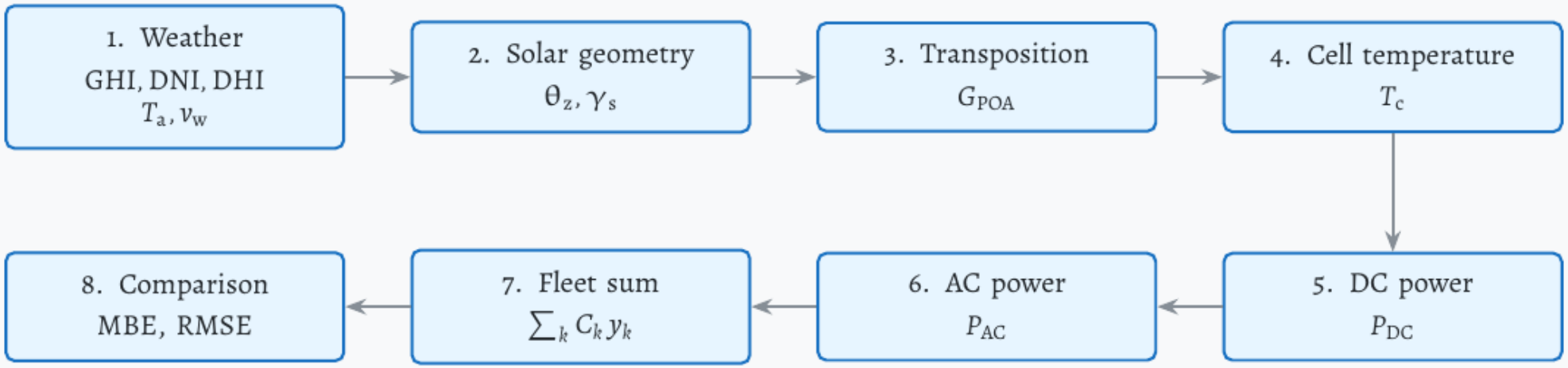
Who am I

PV performance model steps 1 and 2

Notebook 1: Python intro + POA irradiance

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Modeling steps



Modeling steps

1. Weather data

Irradiance, ambient temperature, wind speed, hourly.

Satellite products (PVGIS SARAH3, CAMS), reanalysis (ERA5), ground stations.

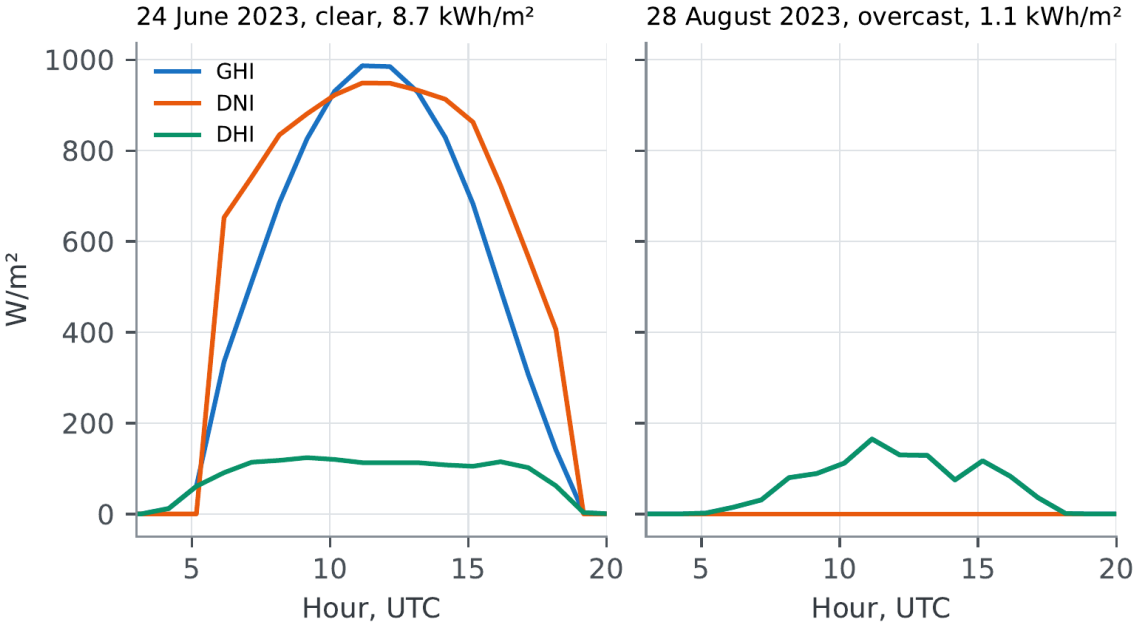
Three irradiance components, W/m²:

- GHI, global horizontal
- DNI, direct normal, measured normal to the rays
- DHI, diffuse horizontal

Clear 24 June: 8.7 kWh/m², diffuse 17 %. Overcast 28 August: 1.1 kWh/m², all diffuse, DNI zero.

Over 2023, diffuse is 41 % of GHI here. DNI stays at zero after sunrise until the sun clears the mountains.

$$\mathrm{GHI} = \mathrm{DNI}\cos\theta_z + \mathrm{DHI}$$



Data: PVGIS SARAH3, course site 45.64° N, 5.92° E, year 2023. PVGIS horizon profile included. Jensen et al. (2023), pvlib iotools, Solar Energy.

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Modeling steps

1. Weather data sources

Source	Type	Step	Note
PVGIS (SARAH3)	Satellite	1 h	Europe and Africa, 2005 onward, free API
PVGIS (ERA5)	Reanalysis	1 h	Global, coarser grid
CAMS radiation	Satellite	1 min to 1 h	Registration needed
Meteo-France	Station	1 h	Ground truth, sparse network

A satellite pixel covers a few kilometres. One rooftop sees a different sky under broken cloud.
The mismatch averages out over a department and a year, and stays visible hour by hour at one site.

Modeling steps

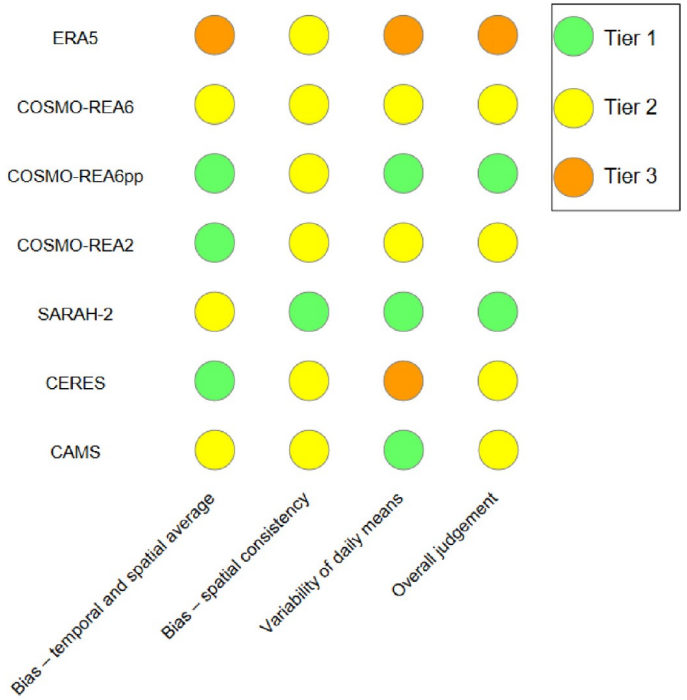
1. Which irradiance product?

Seven gridded products driving the same PV model for Germany, ranked on bias, spatial consistency and day-to-day variability.

ERA5, the reanalysis, gives the largest biases in spring, autumn and winter, and sits in the lowest tier overall.

Satellite products (SARAH-2, CAMS) sit in the middle tiers. PVGIS SARAH3, used in this course, belongs to the SARAH family.

The weather input alone moves the modelled fleet production by several percent.



Kenny, Fiedler (2022), Solar Energy 232, 444–458, doi:10.1016/j.solener.2021.12.044, CC BY 4.0.

Modeling steps

1. Weather data: sun path

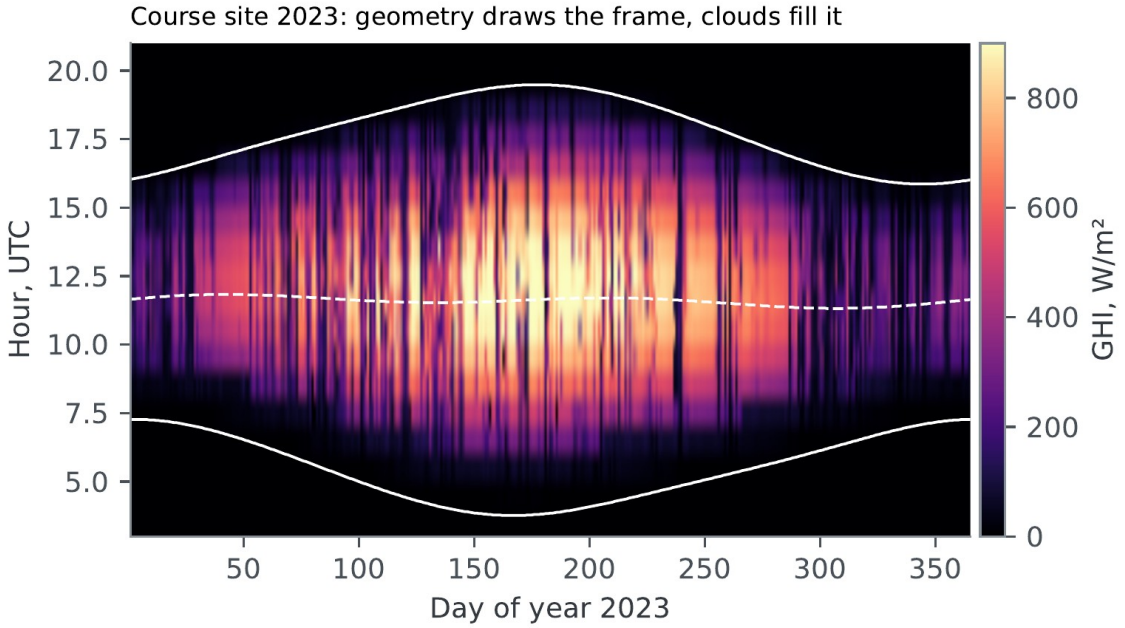
The sun position is characterized by:

- h [°], elevation angle, height of the sun in the sky
- Φ [°], azimuth angle, between north and the sun direction, clockwise
- θz = 90° − h, the zenith angle

Noon elevation = 90° − φ + δ: 67.8° in June, 20.9° in December at 45.6° N.

The figure: every hour of 2023. White lines, computed sunrise, sunset and solar noon. Colour, measured GHI.

Day length 8.7 h to 15.7 h. Solar noon moves between 11:19 and 11:50 UTC over the year: the equation of time.



Data: PVGIS SARAH3, course site 45.64° N, 5.92° E, year 2023. Sunrise and sunset: NREL SPA, Reda and Andreas (2004), in pvlib.

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Modeling steps

1. Solar geometry

δ, declination, day n

ω, hour angle, from the solar time ts

θz, zenith, at latitude φ

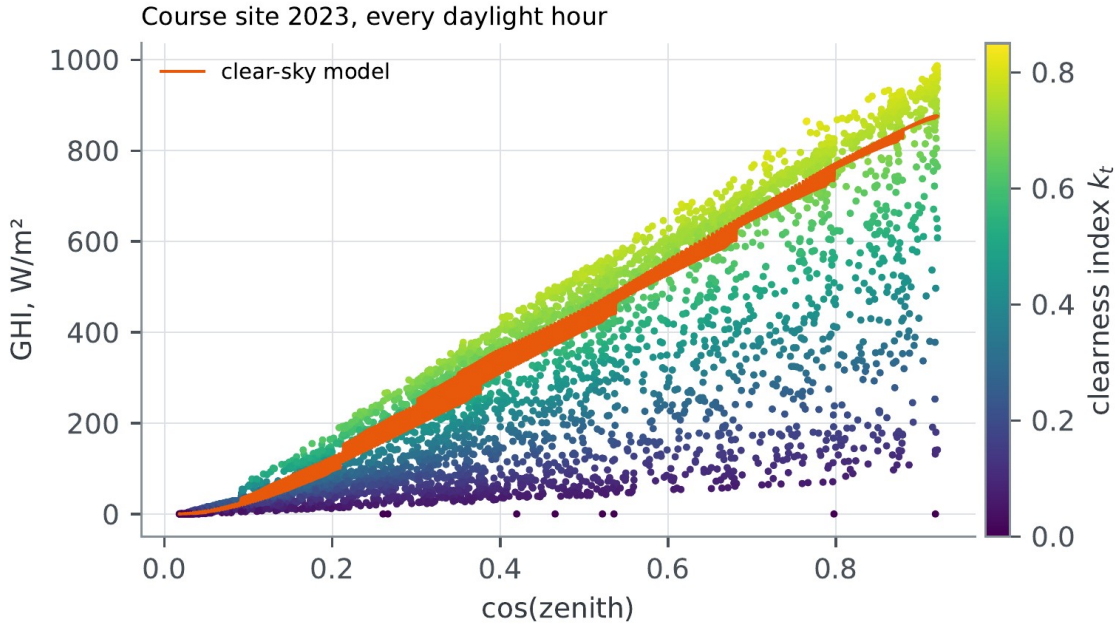
γs, solar azimuth

Solar time differs from clock time: 4 min per degree of longitude, plus the equation of time (−14 to +16 min), plus summer time.

Work in UTC inside the model.

The figure: geometry sets the ceiling, clouds decide the rest. Clear hours sit on the clear-sky line, 23 % of daylight hours. 30 % are overcast, kt below 0.25.

$$\cos\theta_z = \sin\delta\sin\phi + \cos\delta\cos\phi\cos\omega$$



Data: PVGIS SARAH3, course site 45.64° N, 5.92° E, year 2023. Clear sky: Ineichen model. Cooper (1969), Spencer (1971).

Modeling steps

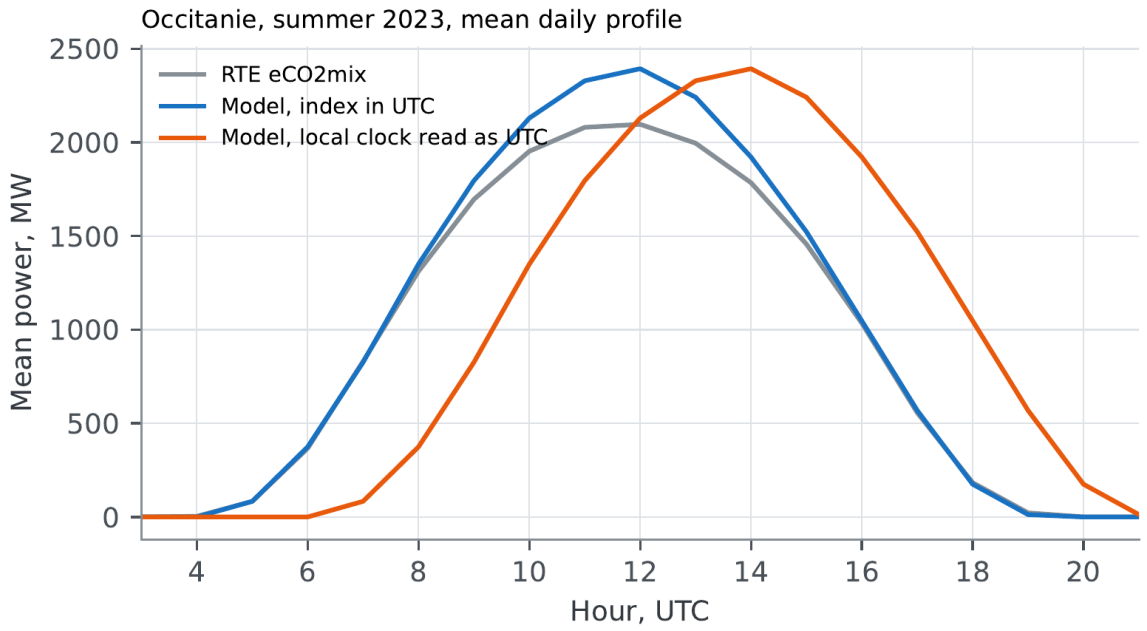
1. Time conventions

Four questions about any irradiance file:

- UTC or local, with or without summer time?
- Instantaneous value or interval mean?
- Label at the start, middle or end of the interval?
- W/m² or Wh/m² per step?

Occitanie 2023: a model read in local clock time instead of UTC drops the hourly R^2 against RTE from 0.96 to 0.61, two hours off in summer.

The annual energy stays identical. Only the profile reveals the error.



Data: RTE eCO2mix Occitanie 2023, PVGIS SARAH3 at 13 department nodes. Model: 30° south, isotropic, flat 20 % losses.

Modeling steps

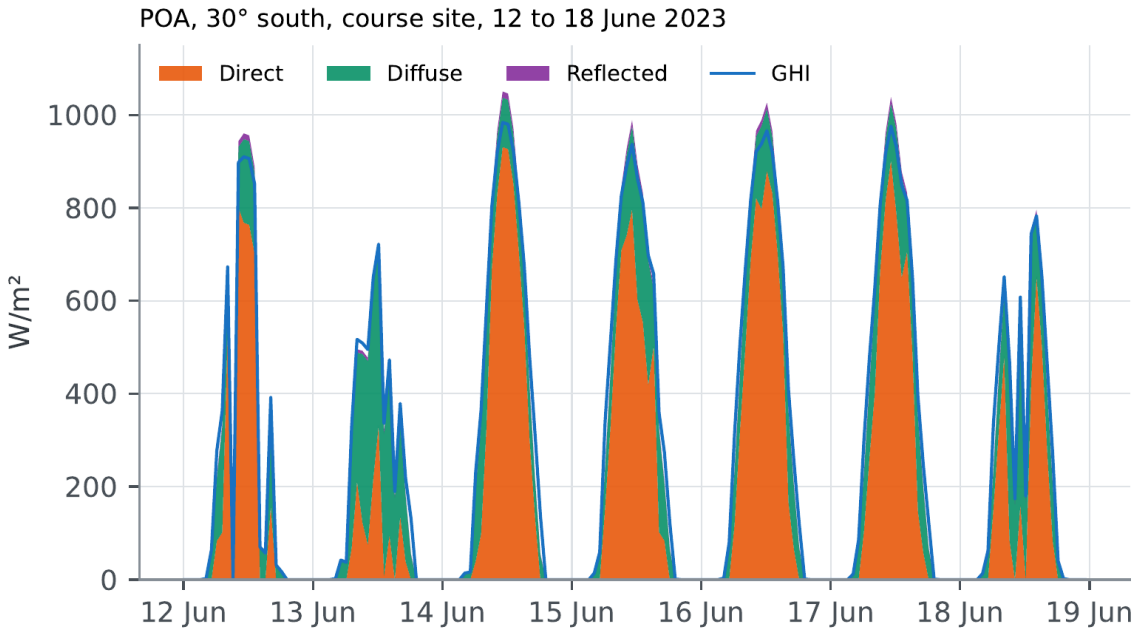
2. Plane-of-array irradiance

The global POA irradiance is calculated from DNI, DHI and GHI and is broken down into 3 components:

- Direct: DNI · cos(AOI)
- Diffuse: from DHI, isotropic model DHI (1 + cos β) / 2
- Reflected: GHI · ρ · (1 − cos β) / 2

With β the tilt, ρ the ground albedo (0.2 grass and urban, 0.6 to 0.9 snow), AOI the angle of incidence.

Course site 2023, 30° south: 1564 kWh/m² on the plane against 1380 on the horizontal, +13 %. Direct 65 %, diffuse 34 %, reflected 1.2 %.



$$G_{POA} = \mathrm{DNI}\cos\theta + D_{poa} + R_{poa}$$

Data: PVGIS SARAH3, course site 45.64° N, 5.92° E, year 2023. Liu and Jordan (1960). pvlib.irradiance.get_total_irradiance

Modeling steps

2. POA: angle of incidence

AOI, the angle between the sun rays and the module normal.

cos(AOI) < 0 means the sun is behind the plane: clip to zero.

Test case: south-facing surface at solar noon, AOI = $|\varphi - \delta - \beta|$.

Azimuth convention in pvlib: clockwise from north, south = 180°.

$$\cos\theta = \cos\theta_z \cos\beta + \sin\theta_z \sin\beta \cos(\gamma_s - \gamma)$$

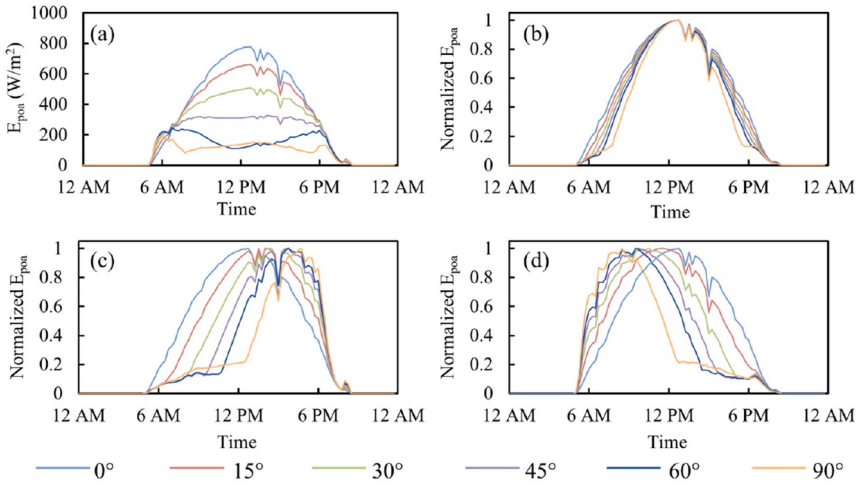


Figure: effect of the tilt on the POA profile, (a) north, (b) south, (c) east, (d) west. Meng, Loonen, Hensen (2020), Solar Energy 211, 418–432, doi:10.1016/j.solener.2020.09.077, CC BY 4.0.

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Modeling steps

2. POA: sky diffuse models

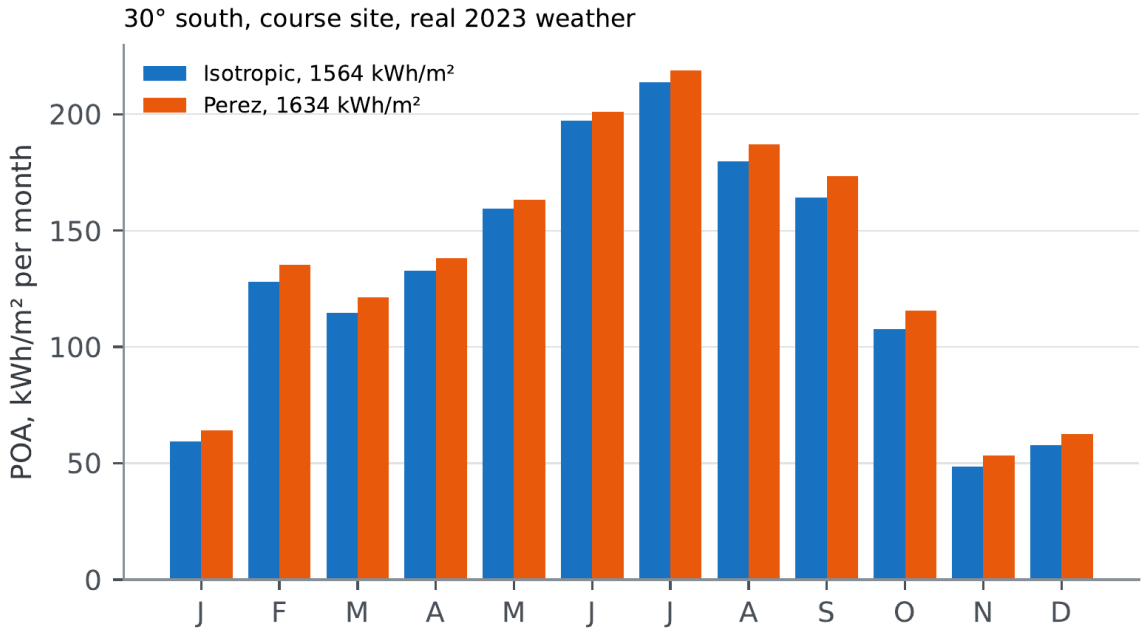
Isotropic: uniform sky dome. One line, no coefficient, exact under overcast sky.

Hay and Davies: a circumsolar part treated as direct, plus an isotropic rest.

Perez: circumsolar disc, horizon band, isotropic dome, coefficients binned by sky clearness and brightness.

Course site 2023, 30° south: isotropic 4.3 % below Perez over the year.

The gap is seasonal: 7.6 to 8.9 % from November to January, about 2 % in June and July. Low sun gives more weight to the circumsolar and horizon terms the isotropic sky ignores.



Data: PVGIS SARAH3, course site 45.64° N, 5.92° E, year 2023. Hay and Davies (1980). Perez et al. (1990), Solar Energy 44(5).

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Notebook 1

(Quick) Python introduction and POA calculation

Google Colab link: https://colab.research.google.com/github/Mohamed-ENNHIRI/ENER958/blob/main/notebooks/python_intro_poa.ipynb

First action: File, Save a copy in Drive.

I. Python introduction: print, operations, numpy, Series and DataFrame, datetime index, plots. Execute the cells, do the exercices.

II. POA calculation: one timestep of 21 June 2023, the explicit way and the lazy way with pvlib.

Then one year from the PVGIS data of the course site, and the Perez model to go further.

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Acknowledgements

The structure of this course, the session format and the notebook format follow the course given by Alexandre Mathieu at the Solar Academy in 2024, “PV failure detection based on operational time series”.

Physics and data: pvlib (Holmgren et al., 2018), PVGIS SARAH3, the national register of production installations (ODRE), RTE eCO2mix, BDPV.

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That’s it

Next session: cell temperature and DC power, then notebook 2. Finish part II of notebook 1 before.